

Week 2 Assignment 2

Q1. A bank wants to predict whether a loan application will be approved or rejected.

Is this a classification or regression problem?

This is a classification problem because the output belongs to categories such as “Approved” or “Rejected”.

Which algorithm among Decision Tree, KNN, and Linear Regression would be most suitable?

Decision Tree would be the most suitable algorithm because it works well for classification tasks and can handle both numerical and categorical data effectively.

Q2. Explain the difference between a feature and a label using a real-world example.

A feature is an input variable used to make predictions, while a label is the output or result being predicted.

Example:

In house price prediction:

- Features: Size of house, number of rooms, location
- Label: Price of the house

Q3. A Decision Tree predicts that a customer will purchase a product.

Explain how the model reaches this prediction.

A Decision Tree reaches a prediction by asking a series of decision-based questions. For example, the

model may check the customer's age, income, and shopping history. Based on the answers, it follows different branches of the tree until it reaches a final prediction such as "Will Purchase".

Q4. Why is KNN considered a supervised learning algorithm?

Can KNN perform classification if labels are not available?

KNN is considered a supervised learning algorithm because it learns using labeled training data.

No, KNN cannot perform classification if labels are not available because it needs labeled examples to determine the class of new data points.

Q5. Suppose a KNN model uses:

$K = 5$

and the nearest neighbors are:

Yes, Yes, No, Yes, No

What will be the prediction and why?

The prediction will be "Yes" because among the 5 nearest neighbors:

- Yes = 3
- No = 2

The majority class is "Yes", so KNN predicts "Yes".

Q6. Explain the equation of Linear Regression:

$$y = mx + b$$

What do x , y , m , and b represent?

In this equation:

- x represents the independent variable or input.
- y represents the dependent variable or predicted output.
- m represents the slope of the line.
- b represents the intercept where the line crosses the y -axis.

Q7. What is correlation?

What does each of the following values indicate?

+1

0

-1

Correlation measures the relationship between two variables.

- +1 indicates a perfect positive correlation.
- 0 indicates no correlation.
- -1 indicates a perfect negative correlation.

Q8. A disease detection model produced the following result:

Actual Condition: Disease Present

Prediction: Healthy

What type of prediction is this?

This is a False Negative prediction.

Why can this be dangerous?

This can be dangerous because the patient actually has the disease but the model predicts they are healthy. This may delay treatment and worsen the patient's condition.

Q9. What is the purpose of a Confusion Matrix?

How is it different from a Correlation Matrix?

A Confusion Matrix is used to evaluate the performance of a classification model by comparing actual and predicted values.

A Correlation Matrix shows the relationship or correlation between multiple variables in a dataset.

Q10. Compare the following algorithms in one or two lines each:

Decision Tree

A Decision Tree makes predictions using a tree-like structure of decisions and conditions.

KNN

KNN classifies data based on the majority class of its nearest neighbors.

SVM

SVM separates data into classes using the best possible boundary called a hyperplane.

Linear Regression

Linear Regression predicts continuous numerical values using a linear relationship between variables.

Neural Network

A Neural Network mimics the human brain using layers of interconnected nodes to learn complex patterns from data.